CSE 318 — Online 4

Section A1

The geometric cooling $T \leftarrow \alpha T$ uses a fixed α, which can be too fast in some regimes and too slow in others. A common improvement is to *adapt* α based on how often moves are being accepted: if too many moves are being accepted, the temperature is too high (cool faster); if very few are being accepted, the temperature is too low (cool slower or hold steady).

What to implement.
Modify your SA loop so that cooling happens *every W iterations* (not every iteration), and the value of α used depends on the acceptance ratio in the last window.

1. Set $W = 5$ or 10 (window size).
2. Every W iterations, compute

$$r \;=\; \frac{\text{accepted moves in the last } W \text{ iterations}}{W}.$$

3. Choose the cooling rate for this cooling step as:
   - If $r > 0.5$: $\alpha = 0.90$ (too many accepts, cool faster).
   - If $r < 0.1$: $\alpha = 0.999$ (too few accepts, cool slower).
   - Otherwise: $\alpha = 0.995$ (default).
4. Update $T \leftarrow \alpha T$ and reset the window's accept counter.
5. Keep the same T_0, $T_{\min}$, iteration budget, neighborhood, and acceptance rule as before.

What to show in the output.

- Run modified SA **5 times** on **at least two** instances. Report **best, average, worst** cost.
- Compare against your original SA (fixed $\alpha = 0.995$). Report best/avg/worst for both.